

Yash Rawal

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Master of Computer Applications (AI & ML) | Microsoft Certified: Azure AI Fundamentals (AI-900)

Cover Letter: AI/ML Engineer — ECS Big Bets & AI 10.0 (Req ID: 2333072)

Optum (UnitedHealth Group) Hyderabad, India September 3, 2026

Dear Optum Hiring Team,

I am writing to express my enthusiastic interest in the AI/ML Engineer position within the ECS Big Bets and AI 10.0 initiatives at Optum (UnitedHealth Group) in Hyderabad (Req ID: 2333072).

As an Applied AI & Machine Learning Engineer currently architecting enterprise agentic platforms serving 1M+ active users across 120+ organizations, my background brings together the exact technical pillars required for Optum's next-generation health optimization: Model Context Protocol (MCP) server development, multi-agentic orchestration (LangChain, LangGraph), hierarchical RAG architectures, and clinical NLP experience.

- **Model Context Protocol (MCP) & Multi-Agent Systems**, Architected and deployed enterprise multi-agent workflows utilizing MCP Server Tool Routing (Tool RAG) and A2A communication, designing a capability contract layer that cut execution overhead by 3x and token consumption by 70%.
- **Clinical NLP & Healthcare AI Ingestion**, Engineered clinical summarization pipelines for 116K+ longitudinal medical records using 4-bit quantized Llama 3, and designed real-time WebSocket streaming Indic ASR with NeMo RNN-T and CTC decoding.
- **Hierarchical RAG & Vector Knowledge Retrieval**, Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a 90%+ Recall@10 retrieval rate.
- **Evaluation Control Planes & Production Guardrails**, Built an automated LLM evaluation and red-teaming control plane to score agent reasoning trajectories, and built production guardrails against prompt injections and data leaks with automated PII redaction.

Delivering care and connecting millions of patients with clinical insights through AI 10.0 requires high-concurrency systems engineered with strict privacy compliance, deterministic tool invocation, and low-latency agentic execution. Having built production agentic platforms at 1M+ scale, contributed upstream to Spring AI (PR #5711), and engineered clinical NLP pipelines, I am eager to help Optum advance health optimization on a global scale.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my technical expertise in MCP, multi-agentic architectures, and clinical AI can contribute to Optum's ECS Big Bets initiatives.

Sincerely,
Yash Rawal